

PHYSICAL AI SYSTEMS: THE CO-DESIGN FRAMEWORK & CURRICULUM ROADMAP

From Disembodied Cloud ML Behind Glass to Autonomous Embodied Agents Governing Matter, Momentum, and Energy

PART I: FOUNDATIONS

The Three Pillars & The Physical AI Co-Design Matrix

CHAPTER 1 The Causal Boundary

The Problem & The Discipline:

Crossing from advisory bits into moving matter.

- Action mutates world state: $A_t \rightarrow W_{t+1} \rightarrow O_{t+1}$
- Proposal–Permission privilege separation
- Master 5x5 Physical AI Co-Design Matrix

CHAPTER 2 The Physical Columns

The "Physical" Half (5 Columns):

Physical laws bounding intelligence in silicon & steel.

- Time (P99 tail latency) & Freshness Wall (Δt_{wall})
- Inertia & Dynamic Stopping Kinematics (d_{stop})
- Inductive Rise (L/R), Joule Heat & UMA Bus QoS

CHAPTER 3 The AI Cognitive Rows

The "AI" Half & Multi-Rate Architecture:

Cognitive functions & 3 temporal execution cadences.

- 5 AI Stages: Perceive, Remember, Reason, Plan, Act
- Classical Robotics vs. Modern Physical AI
- 3 Cadences: System 2 (1 Hz) $\leftrightarrow$ 1.5 (30 Hz) $\leftrightarrow$ 1 (1 kHz)

PART II: THE EMBODIED LIFECYCLE

Conquering the Co-Design Matrix Row by Row (Chapters 4–8)

CHAPTER 4

Row 1: Perceive

Spatial Tokens in SE(3)

Sensory Ingestion Gateway:

- CMOS, ToF, and IMU physics
- Zero-copy DMA ring buffers
- UMA bus QoS arbitration
- ViT / DINOv2 patch tokens
- Hardware PTP exposure strobe

Output: 3D Tokens S_t

CHAPTER 5

Row 2: Remember

World Models & Frame Trees

Latent Metric State:

- Non-generative JEPAs / RSSMs
- Dynamic SE(3) frame trees
- Lie algebra (se3) integration
- Spatial persistence on dropout
- Positional covariance bounds (σ)

Output: State Belief B_t

CHAPTER 6

Row 3: Reason

Goals & Expiring Leases

Semantic Deliberation:

- Multimodal VLMs as System 2
- Expiring Intent Leases (L_{intent})
- Workspace & reach screening
- Grounding open-world prompts
- Zero direct motor write access

Output: Intent Lease L_{intent}

CHAPTER 7

Row 4: Plan

Action Chunks & Splines

Trajectory Decoding:

- Diffusion / ACT action chunking
- Deliberation delay amortization
- Receding-horizon ensembling
- C^2 quintic splines (bounded jerk)
- Atomic state re-anchoring

Output: Action Chunk $A_{t:t+H}$

CHAPTER 8

Row 5: Execute

1 kHz MCU Safety Reflex

Real-Time Permission:

- Bare-metal MCU (malloc=0)
- Control Barrier Functions
- Dynamic stopping invariant
- Hardware STO / gate drivers
- Safe state projection (u^*)

Output: Gate Volts $u(t)$

PART III: INTEGRATION, GOVERNANCE & RELEASE

Whole-System Deployment, Placement, and Defensible Release (Chapters 9–12)

CHAPTER 9 Silicon Placement

Heterogeneous SoC placement and MPU/NPU/MCU memory topology.

- Edge vs. Cloud split & dropouts
- UMA memory bus contention
- Thermal derating ripple

CHAPTER 10 Governance & Ops

Human-machine authority arbitration & continuous fleet telemetry.

- Bumpless control transfer
- Intervention tagging & shift
- Policy endogeneity flywheels

CHAPTER 11 System Assurance

Defensible release cases (CAE) & cross-layer fault injection.

- STPA hazard analysis
- 5-tier qualification ladder
- Live bench fault qualification

CHAPTER 12 Capstone Defense

Whole-system dual-brain defense under live bench fault injection.

- Full 11-artifact dossier
- Live multi-fault injection
- Zero unmitigated crashes